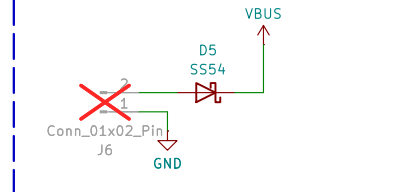
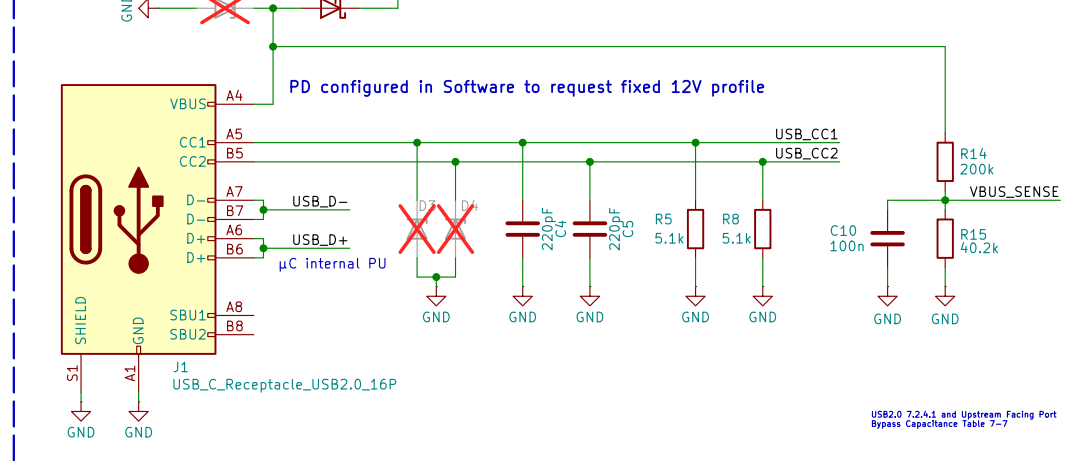


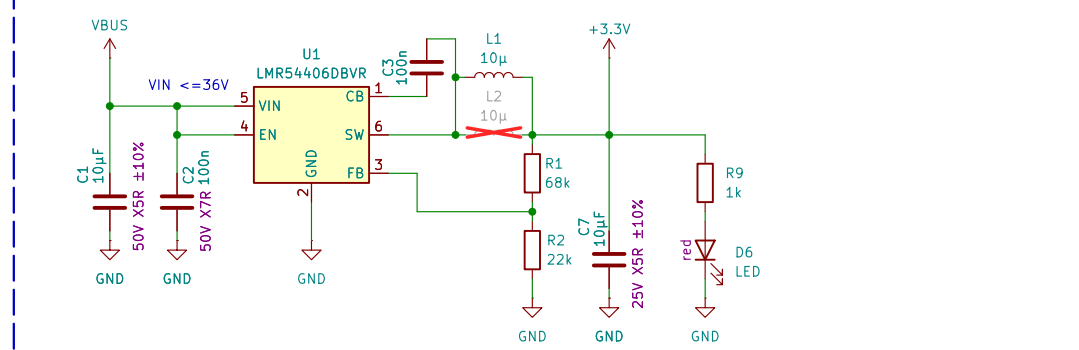
Board Power Supply (+12V)



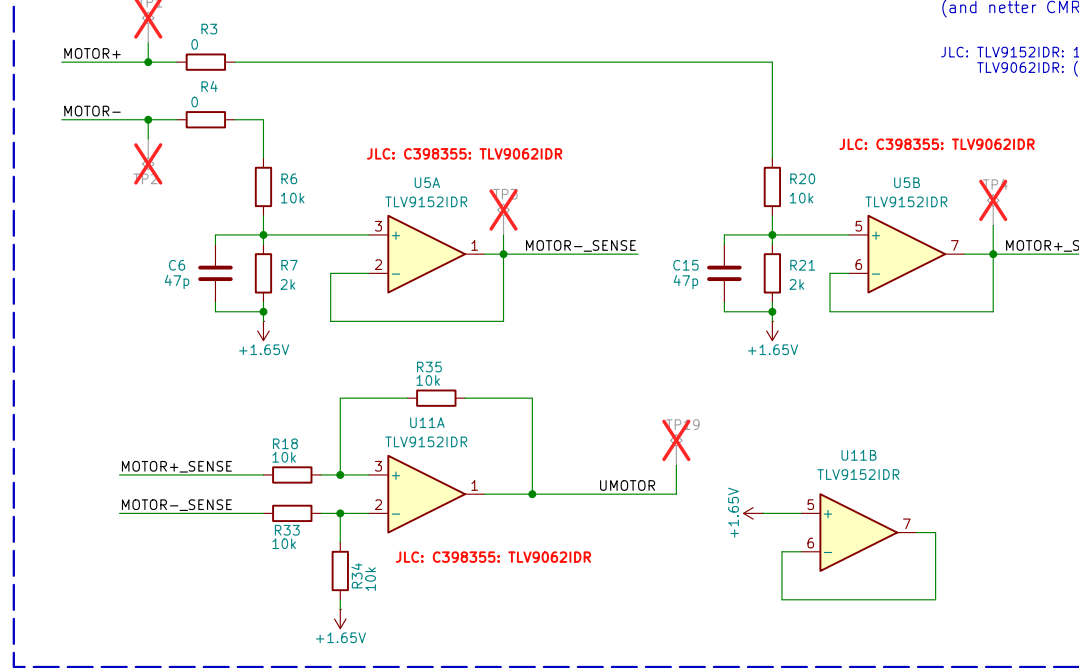
USB Data onboard $\mu C \rightarrow PC$. USBC Power OPTIONAL
NOT used for first experiments
only if we want to power the Board with 9/12V via USBC-PD later
(not sure if usbcpd negotiation has trouble with D2/D5)



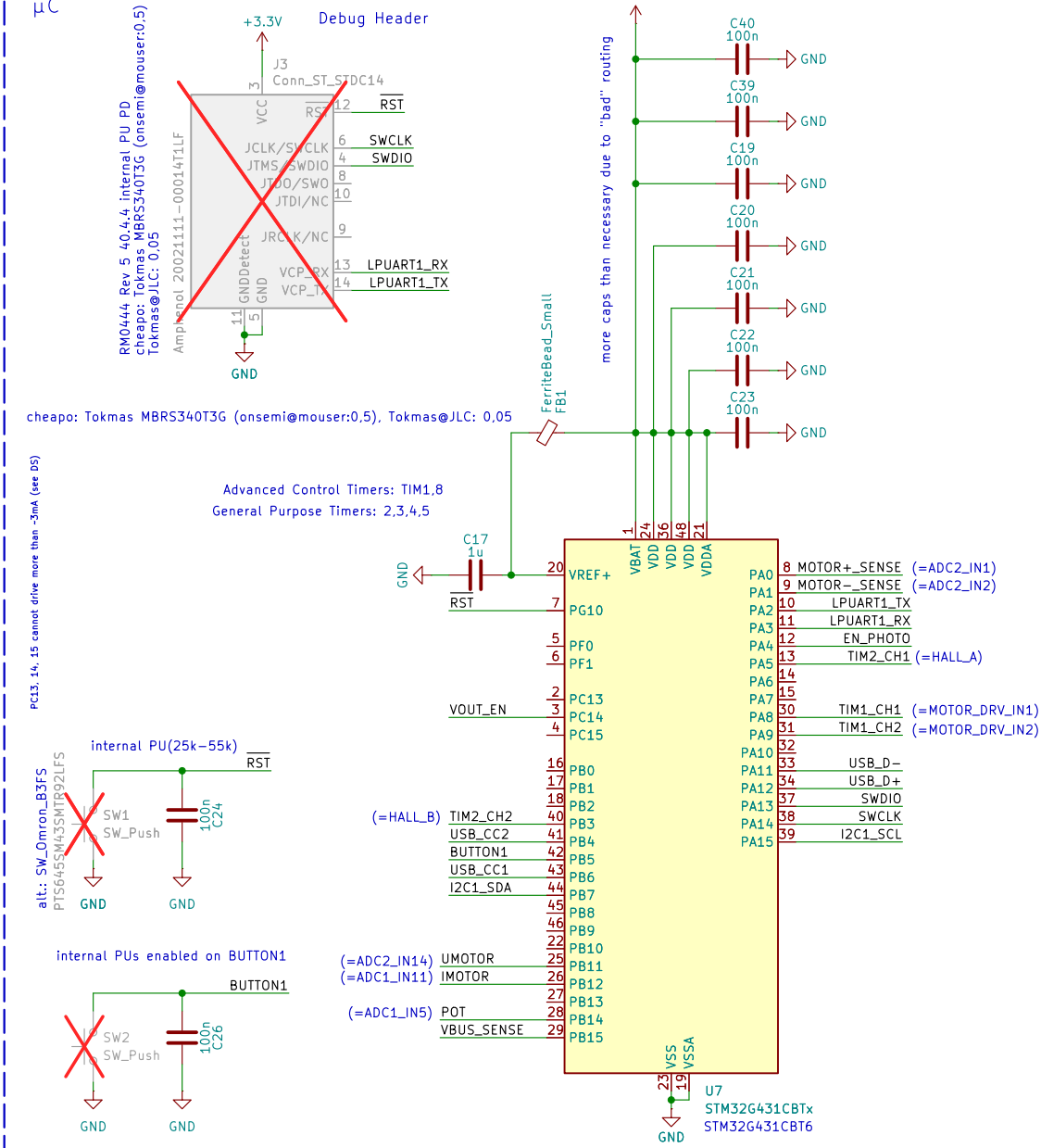
12V $\rightarrow$ 3.3V
-> potentially change to switcher LMR54406BVR
Inductor alternatives: ASP1-40205-100M-T (C3222037), SDCL1V4020-100M-R 74405042100, XAL4040-103ME (C5144883)
handsolder: SPM4020T-100M-LR, ANR4020T100M (C6808085)



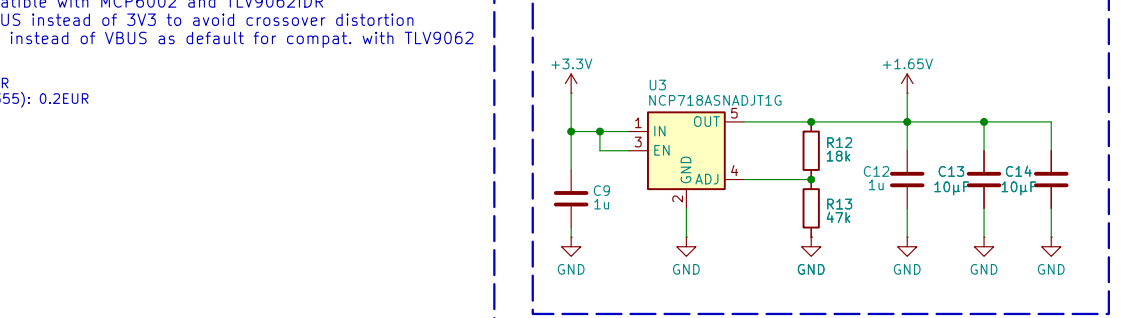
Back-EMF measurement



μC



Virtual GND for Back-EMF Measurement



5V for HALL-Supply and Back-EMF Measurement

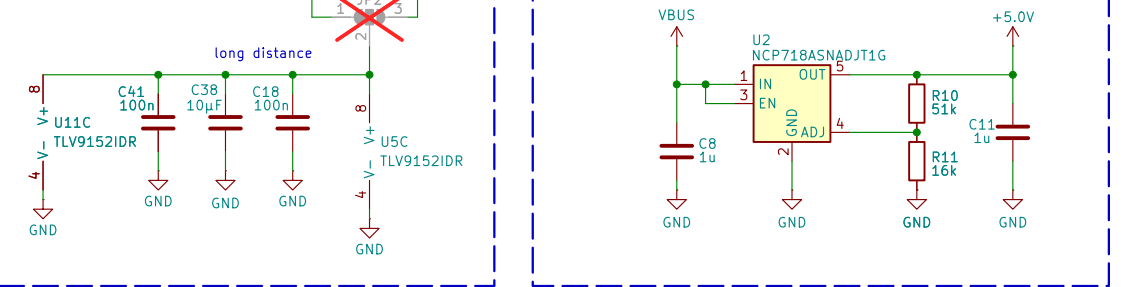
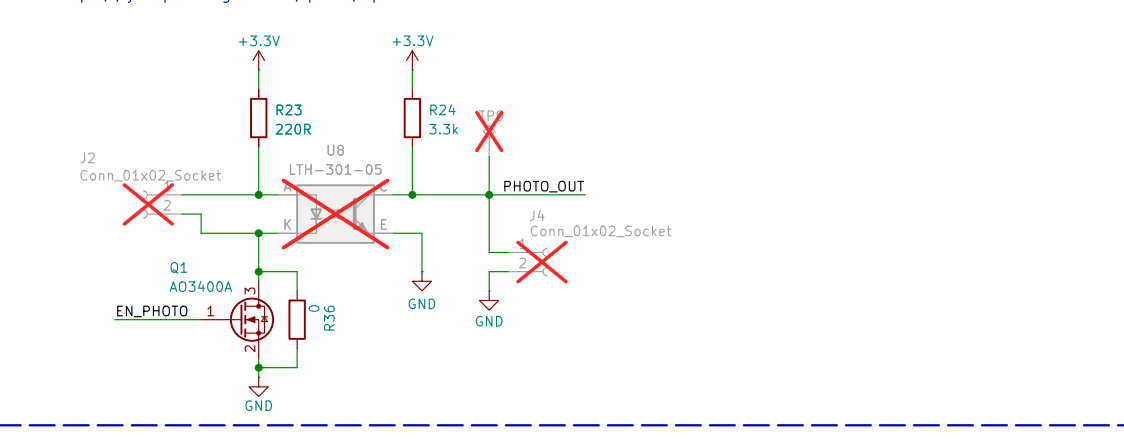
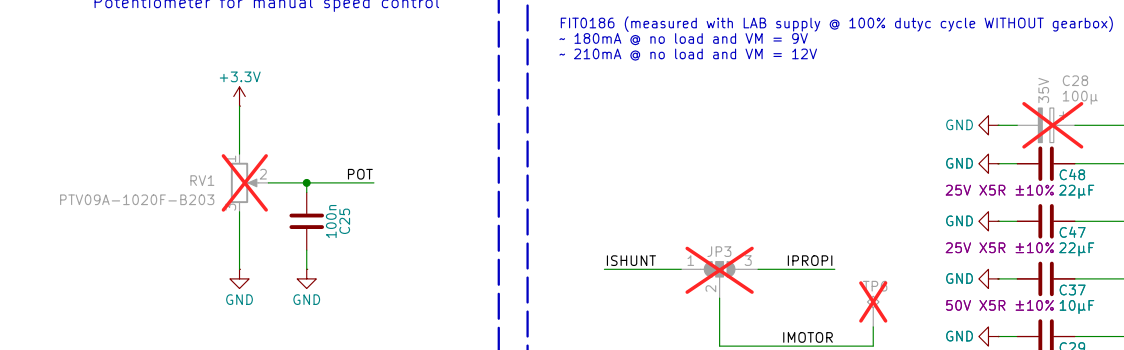


Photo Interruptor

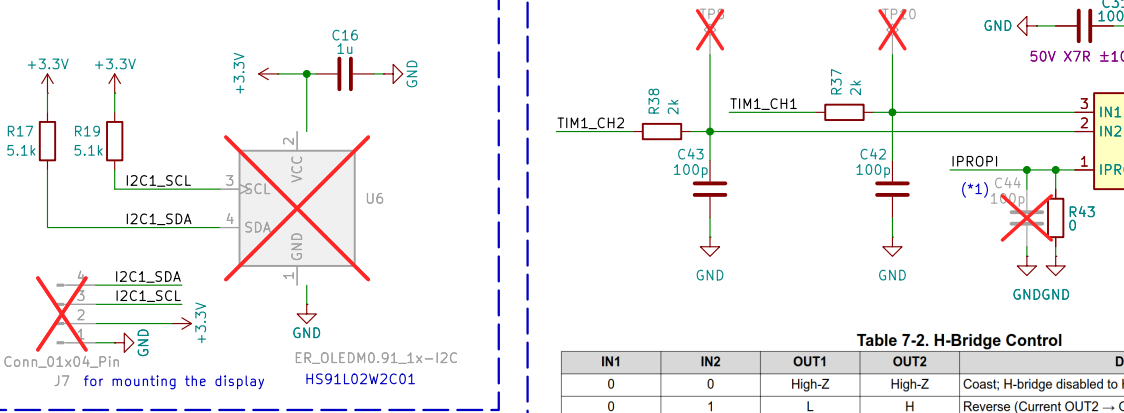
No CTR given in Datasheet
1.)
At $I_F = 20mA \rightarrow$ Datasheet guarantees $> 0.5mA$ photocurrent $\rightarrow$ estimated worst case $CTR \sim 0.5/20 = 2.5\%$
 $\rightarrow$ worst case $I_C @ 10mA$ LED drive $= 240\mu A$
 $\rightarrow$ For $R_{load} = 1k \rightarrow V_{low,out} = 3.3 - 240\mu A * 1k = 3.1V \rightarrow$ only 0.2V swing
2.)
To guarantee saturation in worst case, R_{load} must limit I_C to $240\mu A$
 $R_{load} > (3.3 - V_{CE,sat}) / 240\mu A = 13k\Omega$
3.)
Estimate R_{load} for large enough output swing:
For a swing of 2V $\rightarrow R_{load} = 2 / 240\mu A = 8.3k\Omega$
 $\rightarrow$ we want be rather fast, so try with a smaller R_{load} first
<https://josephehoff.github.io/posts/rpm>



Potentiometer for manual speed control

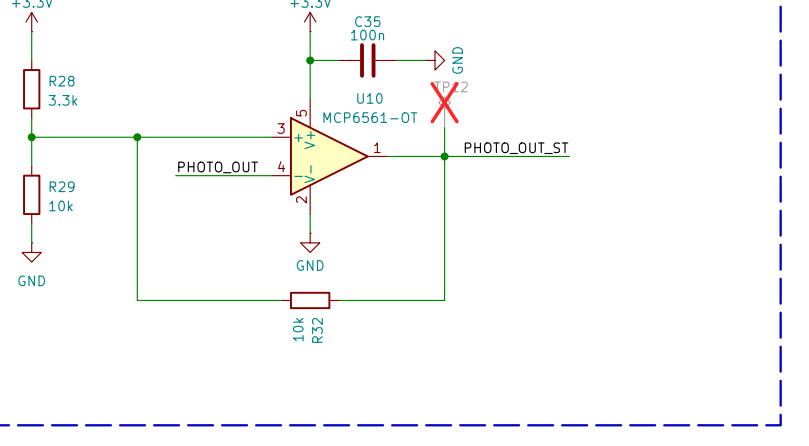


Display (optional)

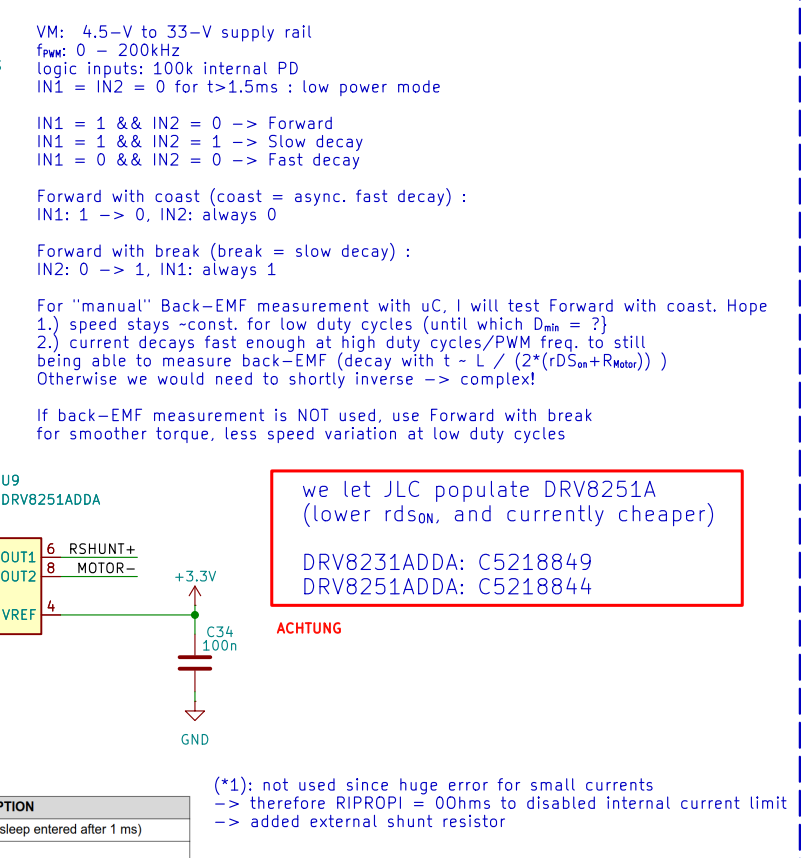


Schmitt-Trigger

Inverting Config
 $\rightarrow$ for Rthevenin $< <$ Feedback : hysteresis-width - independent from threshold center
 $\rightarrow$ source impedance does not influence thresholds
with 1.5V swing from photointerruptor we expect 3.3V (high) $\rightarrow$ 1.8V (low)
 $\rightarrow$ set center threshold to $\sim 2.55V$
 $\rightarrow$ currently: switching threshold: 2.42V, hysteresis: 240mV

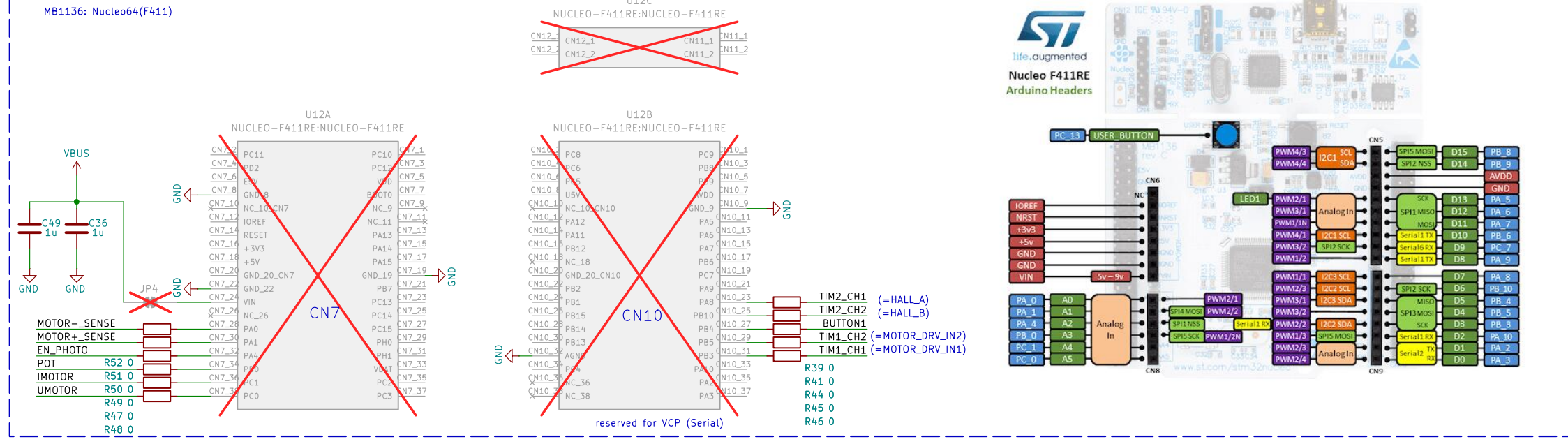


Motor-Driver



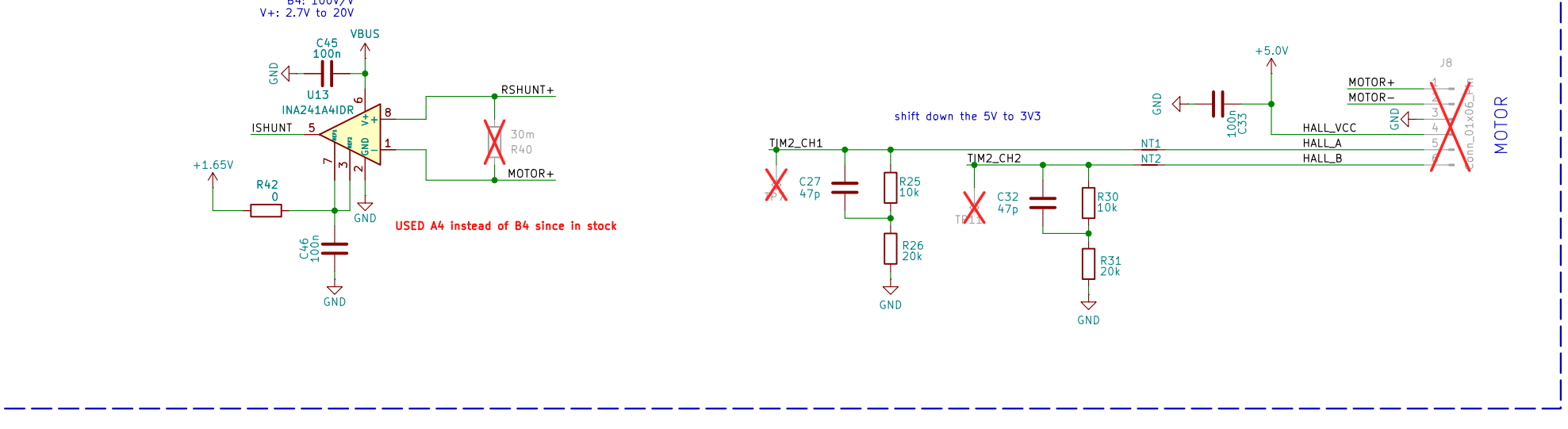
we let JLC populate DRV8251A (lower $r_{DS(on)}$ and currently cheaper)
DRV8231ADDA: C5218849
DRV8251ADDA: C5218844

MATLAB



Motor Connector

pins mapped to allow TIM2 to be configured in CubeMX as "combined channels" in Encoder Mode to measure Motor Hall Encoder Signals as quadrature signals



dummy symbol to include motor mechanical step
Achtung: Für $< 3.3V$: C15850 statt C440198
C1, C29, C30, C37 da für 12V ok und günstiger

DC Motor Driver Board
Elektronik & Technische Informatik
HTL Graz-Gösting
Autor: Franz Patz
Sheet: /
Title: DC Motor Driver Board
Size: A2 Date: 2024-11-30 Rev: V1.0
Kicad E.D.A. 10.0.6 Id: 1/1